Noise-Aware QKD Network Resilience

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ABSTRACT

Noise-aware QKD resilience emerges from combining coexistence engineering, adaptive control, and coding-based protection. Shared-fiber noise lowers reach and key rate, but experiments show viable coexistence in backbone and long-haul links, while adaptive protocols and tailored codes improve robustness under specific noise models.

INTRODUCTION & LITERATURE REVIEW

The practical resilience of quantum key distribution (QKD) depends on managing noise across shared optical infrastructure, quantum channels, and imperfect devices. Coexistence studies demonstrate the feasibility of quantum communication alongside high-capacity classical traffic, including long-distance transmission. [1,2] Security analyses incorporating realistic device imperfections further highlight the importance of implementation-aware protocol design. [3] Complementary advances in noise-adaptive QKD and quantum coding investigate performance improvements under specific channel conditions, including correlated noise and decoherence. [4,5] However, these approaches address different aspects of network performance and are not universally effective across noise models. This motivates a cross-layer perspective that considers coexistence engineering, protocol adaptation, and coding-based protection together to assess their complementary roles in improving QKD network resilience.

METHODOLOGY

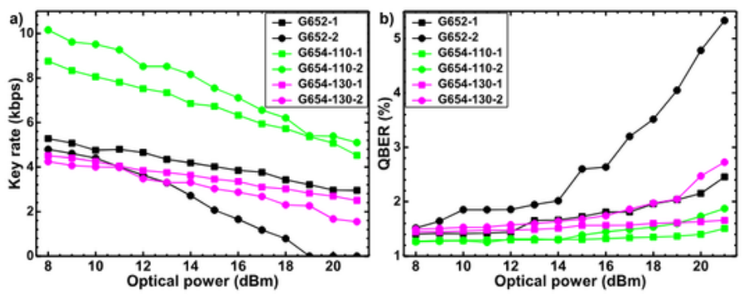
Compare three modes: physical mitigation only, physical plus adaptive control, and physical plus adaptive plus coding-aware networking. Measure secure key rate, QBER or excess noise, outage recovery, and delivered key service under varying launch power and noise models.

FIBER COEXISTENCE

Shared-fiber QKD is feasible, but classical-channel noise sets the main reach and key-rate limit. Backbone and long-haul experiments show coexistence works when filtering, fiber choice, or CV-QKD mode selectivity suppress Raman and related noise, [1,2]

- 3.6 Tb/s backbone traffic coexisted with QKD over 66 km at 21 dBm launch power. [1]
- With 20 GHz filtering and large-core fiber, secure key rate reached 4.5-5.1 kbps in co- and counter-propagation. [1]
- CV-QKD coexisted with fully populated CWDM over 120 km asymptotic and 100 km finite-size without added filtering. [2]

The power-dependent key-rate and QBER tradeoff is plotted directly for shared-fiber backbone coexistence.



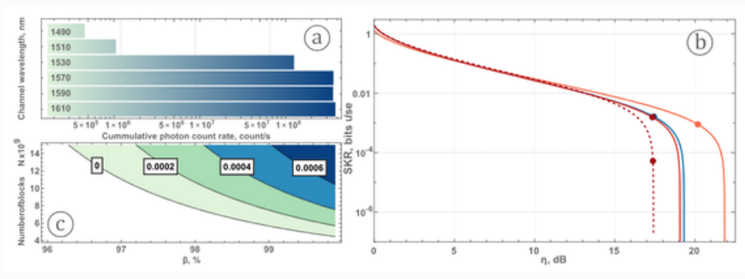
Secure key rate and QBER versus classical power. [1]

ADAPTIVE NOISE CONTROL

Adaptive control addresses time-varying decoherence that fixed QKD settings cannot track. The strongest evidence shows adaptive encoding and decoding improve secure key rate for some noise classes, but not all, so adaptation must be noise-model aware. [3]

- Adaptive two-way QKD improved secure key rates under correlated, non-Markovian, and amplitude-damping-like noise. [3]
- Adaptive strategies showed no benefit for depolarizing and bit-flip Pauli channels. [3]
- CV-QKD coexistence used local-oscillator mode filtering and phase-noise mitigation to sustain long-haul operation. [2]

The long-haul CV-QKD setup and loss-dependent secure key rate under classical coexistence are shown here.



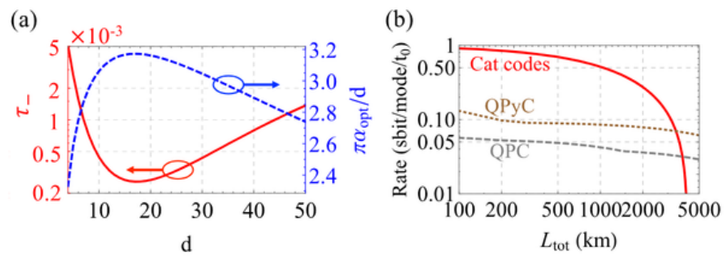
CV-QKD setup and key rate versus channel loss. [2]

CODING-BASED RESILIENCE

Coding papers add a third resilience layer by preserving information under decoherence and loss, although this evidence is more theoretical than network-deployed. Network coding reduces decoherence effects in butterfly-style quantum networks, while bosonic cat codes suppress loss-induced logical errors and can raise communication efficiency per mode, [4,5]

- Quantum network coding transmitted mutual information while reducing decoherence effects and resource cost, [4]
- Cat codes reduced dephasing errors through optimized logical subspace and coherent amplitude, [5]
- One-way repeaters with cat codes achieved higher secure rate per mode than conventional encodings in modeled long-distance links, [5]

Comparison figure shows how optimized coding improves long-distance secure communication rate per mode.



Optimized cat-code performance versus conventional repeater encodings. [5]

DISCUSSION & CONCLUSION

The hypothesis is supported at the component level. Coexistence experiments are strong, adaptive gains are conditional on the noise model, and coding results are promising but less deployment-ready.

REFERENCES

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